ACUTE CARE SURGERY MU HEALTH
ANTIBIOTIC SURGICAL PROPHYLAXIS AND EMPIRIC TREATMENT
PRACTICE MANAGEMENT GUIDELINE

Acute Care Surgery at the University of Missouri is committed to delivering safe, evidence-based, and cost-conscious care across the continuum of trauma, emergency general surgery, critical care, and elective surgical practice. As antimicrobial resistance and practice variation remain ongoing challenges, a unified antibiotic stewardship framework is essential to optimizing patient-centered outcomes and preserving the effectiveness of critical therapies.

This Practice Management Guideline incorporates the principles of the MUHC Adult Surgical Antibiotic Prophylaxis Guidelines and provides diagnosis-specific recommendations informed by contemporary national guidance, specialty-specific literature, local microbiologic trends, and our institutional antibiogram. The Division of Acute Care Surgery has worked collaboratively with the institution's Department of Infectious Diseases and Antimicrobial Stewardship Program to align these recommendations with hospital standards while accounting for the clinical nuances of trauma, emergency general surgery, and critically ill patients.

Consistent with MUHC guidance, surgical antibiotic prophylaxis is one component of a broader strategy to prevent surgical site infection. Appropriate antimicrobial use must complement infection-prevention practices and perioperative optimization, with attention to the anticipated pathogens, patient-specific risks, and potential harms of therapy. This PMG distinguishes prophylaxis from treatment of established infection and recognizes that active infection, prior colonization, multidrug-resistant organisms, and other unique circumstances may warrant individualized management and consultation with Infectious Diseases or the Antimicrobial Stewardship Program.

These recommendations apply to adult patients under the care of the Acute Care Surgery team. By establishing consistent expectations for antibiotic selection, timing, dosing, and duration, this PMG seeks to reduce unnecessary variation, promote judicious antimicrobial use, and align divisional practice with institutional recommendations while preserving sound clinical judgment. Our shared priority is safe, appropriate care directed toward meaningful, patient-centered outcomes.

Acute Care Surgery MU HEALTH Antibiotic Practice Management Guideline: Surgical Prophylaxis

Trauma: Antibiotic Surgical Prophylaxis

Injury/Procedure	Preferred Antibiotic(s)	Alternative Antibiotic(s) if Allergy to Preferred	Duration of Post-Operative Antibiotic(s)
Abdominal Trauma with or without Hollow Viscus Injury	Cefazolin + Metronidazole	Levofloxacin + Metronidazole	24 hours
Chest Tube Placement	N/A	N/A	N/A
Craniotomy	Cefazolin	Vancomycin	24 hours
Open Skull Fracture	Cefazolin	Levofloxacin	24 hours after closure or 72 hours total, whichever is shortest
Open Skull Fracture with Significant Contamination	Cefazolin + Metronidazole	Levofloxacin + Metronidazole	24 hours after closure or 72 hours total, whichever is shortest
Penetrating Brain/CNS Injury	Cefazolin	Levofloxacin	24 hours after closure or 72 hours total, whichever is shortest
Penetrating Brain/CNS Injury with Significant Contamination	Ceftriaxone + Metronidazole	Levofloxacin + Metronidazole	24 hours after closure or 72 hours total, whichever is shortest
Ventriculostomy or ICP Monitor	N/A	N/A	N/A
Pneumocephaly	N/A	N/A	N/A
CSF Leak	N/A	N/A	N/A
Open Face/Mandible Fracture	Cefazolin + Metronidazole	Doxycycline	24 hours after closure or 72 hours total, whichever is shortest
Closed or Non-Operative Face Fracture	N/A	N/A	N/A
Genitourinary Trauma	Cefazolin	Levofloxacin	24 hours
Vascular Trauma	Cefazolin	Vancomycin	24 hours

Emergency General Surgery: Antibiotic Surgical Prophylaxis

Procedure	Preferred Antibiotic(s)	Alternative Antibiotic(s) if Allergy to Preferred	Duration of Post-Operative Antibiotic(s)
Appendectomy	Ceftriaxone + Metronidazole	Levofloxacin + Metronidazole	Non-perforated: 24 hours Perforated with source control: 4 days Perforated without source control 8 days
Colectomy or Bowel Resection	Cefazolin + Metronidazole	Levofloxacin + Metronidazole	N/A
Small Bowel Obstruction with Necrosis or Bowel Resection	Piperacillin-tazobactam	Levofloxacin + Metronidazole	4 days after source control or up to 8 days in critical illness
Small Bowel Obstruction without Necrosis or Bowel Resection	Cefazolin + Metronidazole	Levofloxacin + Metronidazole	24 hours
Perforated Peptic Ulcer Disease Patch Repair	Piperacillin-tazobactam	Levofloxacin + Metronidazole	4 days after source control or up to 8 days in critical illness
Necrotizing Soft Tissue Infection Debridement	Linezolid + Piperacillin-tazobactam	Linezolid alternative: Clindamycin + Vancomycin Piperacillin-tazobactam alternative: Cefepime + Metronidazole	48-72 hours after final debridement, clinical improvement, and fever has been absent

Elective Surgery: Antibiotic Surgical Prophylaxis			
Injury/Procedure	**Preferred Antibiotic(s)**	**Alternative Antibiotic(s) if Allergy to Preferred**	**Duration of Post-Operative Antibiotic(s)**
Cholecystectomy	Cefazolin	Vancomycin	N/A
Inguinal Hernia	Cefazolin	Vancomycin	N/A
Ventral Hernia	Cefazolin	Vancomycin	N/A
Colostomy/Ileostomy Reversal	Cefazolin + Metronidazole	Vancomycin + Metronidazole	N/A
Cardiac & Thoracic	Cefazolin	Vancomycin	N/A
Skin & Soft Tissue	Cefazolin	Vancomycin	N/A

Intra-operative re-dosing:

- Re-dosing of antibiotic intra-operatively is indicated if the duration of surgery exceeds 2 half-lives of the antibiotic or for every 1,500 mL (or >25ml/kg or > 30% blood volume loss for patients < 40kg) of blood loss (regardless of the agent used)
- For prolonged surgeries, re-dose the antibiotic(s) at the indicated redosing interval

For patients already receiving antibiotic treatment before surgery:

- If treatment antibiotic is the **SAME** as the preferred antibiotic for surgical prophylaxis, re-dose antibiotic prior to incision if time since last dose exceeds intraoperative re-dosing interval listed
- If the treatment antibiotic is **DIFFERENT** from the preferred antibiotic for surgical prophylaxis, then administer the preferred pre-operative antibiotic as listed for the procedure, within the appropriate administration window

Dosing and Redosing of Intravenous Antibiotics for Surgical Prophylaxis			
Antibiotic	**Dose**	**Redosing Interval for OR Only (hours)**	**Infusion Time for OR Only (min)**
Ampicillin-sulbactam	3 g	2	30
Cefazolin	2 g if < 120 kg, 3 g if ≥ 120 kg	3	IV push 3-5
Ceftriaxone	2 g	12	IV push 4-5
Ciprofloxacin	400 mg	8	60
Levofloxacin	750 mg	Not recommended	90
Metronidazole	500 mg	8	30
Piperacillin-tazobactam	4.5 g	2	30
Vancomycin	15-20 mg/kg	Not recommended	60 min per 1000 mg

Acute Care Surgery MU HEALTH Antibiotic Practice Management Guideline: Empiric Antibiotic Therapy

Empiric Antimicrobial Therapy for Acute Infections

Acute Infection	Preferred Empiric Antibiotic(s)	Alternative Antibiotic(s) if Allergy to Preferred	Duration of Antibiotic(s)
Community Acquired Pneumonia (CAP)	Amoxicillin-clavulanate + Azithromycin	Amoxicillin-clavulanate alternative: Ceftriaxone Azithromycin alternative: Doxycycline	5 days Azithromycin: 3 days
Hospital Acquired or Ventilator Associated Pneumonia (HAP/VAP)	Vancomycin + Piperacillin-tazobactam	Vancomycin alternative: Linezolid Piperacillin-tazobactam alternative: Cefepime	7 days
Diverticulitis with Source Control	Piperacillin-tazobactam	Levofloxacin + Metronidazole	4 days after source control or up to 8 days in critical illness
Diverticulitis without Source Control	Piperacillin-tazobactam	Levofloxacin + Metronidazole	8 days
Cholecystitis	Piperacillin-tazobactam	Levofloxacin + Metronidazole	Post-Cholecystectomy with local inflammation: 24 hours Perforated with source control (includes PCT): 4 days Perforated without source control: 8 days
Bacteremia (Gram Positive Organism)	Vancomycin	Linezolid	Variable based on organism and patient specific factors
Bacteremia (Gram Negative Organism)	Piperacillin-tazobactam	Cefepime	Variable based on organism and patient specific factors
Fungemia	Micafungin	N/A	Variable based on organism and patient specific factors
Complicated Urinary Tract Infection (Pyelonephritis)	Ceftriaxone	Sulfamethoxazole-trimethoprim or Ciprofloxacin	7 days
Uncomplicated Urinary Tract Infection (Cystitis)	Nitrofurantoin	Sulfamethoxazole-trimethoprim or Fosfomycin	Females: 3 days Males: 5 days
Sinusitis	Ampicillin-sulbactam	Levofloxacin + Metronidazole	5 days
Cellulitis - Purulent (or MRSA risk factors)	Vancomycin	Doxycycline or Sulfamethoxazole-trimethoprim	5-10 days
Cellulitis - Non-Purulent	Cefazolin	Cephalexin or Sulfamethoxazole-trimethoprim	5 days
Necrotizing Soft Tissue Infection	Linezolid + Piperacillin-tazobactam	Linezolid alternative: Clindamycin + Vancomycin Piperacillin-tazobactam alternative: Cefepime + Metronidazole	48-72 hours after final debridement, clinical improvement, and fever has been absent

Dosing and Frequency of Empiric Antimicrobials for Acute Infections

Antibiotic	Dose	Route	Frequency	Antibiotic	Dose	Route	Frequency
Amoxicillin-clavulanate	875 mg / 125 mg	PO/per tube	Q12H	Fosfomycin	3 g	PO/per tube	Once
Ampicillin-sulbactam	3 g	IV	Q6H	Levofloxacin	750 mg	IV or PO/per tube	Q24H
Azithromycin	500 mg	IV or PO/per tube	Q24H	Linezolid	600 mg	IV or PO/per tube	Q12H
Cefazolin	2 g	IV	Q8H	Metronidazole	500 mg	IV or PO/per tube	Q8H or Q12H
Cefepime	2 g	IV	Q8H	Micafungin	100 mg	IV	Q24H
Ceftriaxone	2 g	IV	Q24H	Nitrofurantoin	100 mg	PO/per tube	Q12H
Cephalexin	500 mg	PO/per tube	Q6H	Piperacillin-tazobactam	4.5 g	IV	Q8H
Ciprofloxacin	400 mg (IV), 500 mg (PO)	IV or PO/per tube	Q12H	Sulfamethoxazole-trimethoprim	800 mg / 160 mg	PO/per tube	Q12H
Clindamycin	900 mg	IV	Q8H	Vancomycin	Pharmacy to dose	IV	Pharmacy to dose
Doxycycline	100 mg	IV or PO/per tube	Q12H	-	-	-	-

The antimicrobial dosing and frequencies listed above is standard dosing for adult patients with normal renal function; Dose adjustments may be necessary in specific patient scenarios

Musculoskeletal Open Extremity Fractures

Open Extremity Fracture Classification:

Gustilo-Anderson Classification	Open Fracture Description
Type I	Open fracture with a wound < 1 cm long, low energy, without gross contamination
Type II	Open fracture with a wound 1-10 cm long, low energy, without gross contamination or extensive soft-tissue damage, flaps, or avulsions
Type III	IIIA: Open fracture with a wound greater than 10 cm with adequate soft-tissue coverage, or any open fracture due to high-energy trauma or with gross contamination, regardless of the size of the wound IIIB: Open fracture with extensive soft-tissue injury or loss, with periosteal stripping and bone exposure that requires soft-tissue coverage in the form of muscle rotation or transfer IIIC: Open fracture associated with arterial injury ***requiring repair***

Antibiotic Timing:

- Administer within 30 min of arrival to the ED & all patients to have MRSA nasal screen

Antibiotic Selection:

- Type I & Type II Fractures – **Cefazolin** 2 g Q8H
  - If penicillin allergy – **Vancomycin**: pharmacy to dose
- Type III Fractures – **Ceftriaxone** 2 g Q24H + **Vancomycin**: pharmacy to dose
  - If penicillin allergy – **Levofloxacin** 500 mg Q24H

Open Fracture Contamination: Applies to type I, II, & III fractures

- Soil, fecal matter, standing water contamination – **Cefepime** 2 g Q8H + **Metronidazol**e 500 mg Q12H + **Vancomycin**: pharmacy to dose
  - If penicillin allergy – **Levofloxacin** 500 mg Q24H + **Metronidazol**e 500 mg Q12H

Antibiotic Duration:

- Type I & Type II Fractures – **24 hours total**
- Type III Fractures – **24 hours after closure** or **72 hours total**, whichever duration is shorter

Reference

1. Anderson, P. A., Savage, J. W., Vaccaro, A. R., Radcliff, K., Arnold, P. M., Lawrence, B. D., & Shamji, M. F. (2017). Prevention of surgical site infection in spine surgery. *Neurosurgery, 80*(3S), S114–S123. https://doi.org/10.1093/neuros/nyw066
2. Appelbaum, R. D., Farrell, M. S., Gelbard, R. B., Hoth, J. J., Jawa, R. S., Kirsch, J. M., Mandell, S., Nohra, E. A., Rinderknecht, T., Rowell, S., Cuschieri, J., & Stein, D. M. (2024). Antibiotic prophylaxis in injury: An American Association for the Surgery of Trauma Critical Care Committee clinical consensus document. *Trauma Surgery & Acute Care Open, 9*(1), Article e001304. https://doi.org/10.1136/tsaco-2023-001304
3. Ban, K. A., Minei, J. P., Laronga, C., Harbrecht, B. G., Jensen, E. H., Fry, D. E., Itani, K. M. F., Dellinger, E. P., Ko, C. Y., & Duane, T. M. (2017). American College of Surgeons and Surgical Infection Society: Surgical site infection guidelines, 2016 update. *Journal of the American College of Surgeons, 224*(1), 59–74. https://doi.org/10.1016/j.jamcollsurg.2016.10.029
4. Berríos-Torres, S. I., Umscheid, C. A., Bratzler, D. W., Leas, B., Stone, E. C., Kelz, R. R., Reinke, C. E., Morgan, S., Solomkin, J. S., Mazuski, J. E., Dellinger, E. P., Itani, K. M. F., Berbari, E. F., Segreti, J., Parvizi, J., Blanchard, J., Allen, G., Kluytmans, J. A. J. W., Donlan, R., & Schecter, W. P. (2017). Centers for Disease Control and Prevention guideline for the prevention of surgical site infection, 2017. *JAMA Surgery, 152*(8), 784–791. https://doi.org/10.1001/jamasurg.2017.0904
5. Branch-Elliman, W., O'Brien, W., Strymish, J., Itani, K., Wyatt, C., & Gupta, K. (2019). Association of duration and type of surgical prophylaxis with antimicrobial-associated adverse events. *JAMA Surgery, 154*(7), 590–598. https://doi.org/10.1001/jamasurg.2019.0569
6. Bratzler, D. W., Dellinger, E. P., Olsen, K. M., Perl, T. M., Auwaerter, P. G., Bolon, M. K., Fish, D. N., Napolitano, L. M., Sawyer, R. G., Slain, D., Steinberg, J. P., & Weinstein, R. A. (2013). Clinical practice guidelines for antimicrobial prophylaxis in surgery. *American Journal of Health-System Pharmacy, 70*(3), 195–283. https://doi.org/10.2146/ajhp120568
7. Bultas, A. C., Bery, A. I., Deal, E. N., Hartmann, A. P., Richter, S. K., & Call, W. B. (2019). Predictors of treatment failure following de-escalation to a fluoroquinolone in culture-negative nosocomial pneumonia. *Annals of Pharmacotherapy, 53*(12), 1207–1213. https://doi.org/10.1177/1060028019867114
8. Chauv, S., Fontaine, G. V., Hoang, Q. P., McKinney, C. B., Baldwin, M., Buckel, W. R., Collingridge, D. S., Majercik, S., & Wohlt, P. D. (2016). Risk of resistant organisms and *Clostridium difficile* with prolonged systemic antibiotic prophylaxis for central nervous system devices. *Neurocritical Care, 25*(1), 128–132. https://doi.org/10.1007/s12028-016-0254-x
9. Coccolini, F., Sartelli, M., Sawyer, R., Rasa, K., Ceresoli, M., Viaggi, B., Catena, F., Damaskos, D., Cicuttin, E., Cremonini, C., Moore, E. E., Biffl, W. L., & Coimbra, R. (2024). Antibiotic prophylaxis in trauma: Global Alliance for Infection in Surgery, Surgical Infection Society Europe, World Surgical Infection Society, American Association for the Surgery of Trauma, and World Society of Emergency Surgery guidelines. *Journal of Trauma and Acute Care Surgery, 96*(4), 674–682. https://doi.org/10.1097/TA.0000000000004233
10. Corsini Campioli, C., Challener, D., Comba, I. Y., Shah, A., Wilson, W. R., Sohail, M. R., Van Gompel, J. J., & O'Horo, J. C. (2022). Overview and risk factors for postcraniotomy surgical site infection: A four-year experience. *Antimicrobial Stewardship & Healthcare Epidemiology, 2*(1), Article e14. https://doi.org/10.1017/ash.2021.258
11. De Bus, L., Depuydt, P., Steen, J., Dhaese, S., De Smet, K., Tabah, A., Akova, M., Cotta, M. O., De Pascale, G., Dimopoulos, G., Fujitani, S., Garnacho-Montero, J., Leone, M., Lipman, J., Ostermann, M., Paiva, J.-A., Schouten, J., Sjövall, F., Timsit, J.-F., . . . De Waele, J. J. (2020). Antimicrobial de-escalation in the critically ill patient and assessment of clinical cure: The DIANA study. *Intensive Care Medicine, 46*(7), 1404–1417. https://doi.org/10.1007/s00134-020-06111-5

12. Fagon, J.-Y., Chastre, J., Wolff, M., Gervais, C., Parer-Aubas, S., Stéphan, F., Similowski, T., Mercat, A., Diehl, J.-L., Sollet, J.-P., & Tenaillon, A. (2000). Invasive and noninvasive strategies for management of suspected ventilator-associated pneumonia: A randomized trial. *Annals of Internal Medicine, 132*(8), 621–630. https://doi.org/10.7326/0003-4819-132-8-200004180-00004
13. Farrell, M. S., Agapian, J. V., Appelbaum, R. D., Filiberto, D. M., Gelbard, R., Hoth, J., Jawa, R., Kirsch, J., Kutcher, M. E., Nohra, E., Pathak, A., Paul, J., Robinson, B., Cuschieri, J., & Stein, D. M. (2024). Surgical and procedural antibiotic prophylaxis in the surgical ICU: An American Association for the Surgery of Trauma Critical Care Committee clinical consensus document. *Trauma Surgery & Acute Care Open, 9*(1), Article e001305. https://doi.org/10.1136/tsaco-2023-001305
14. Fried, H. I., Nathan, B. R., Rowe, A. S., Zabramski, J. M., Andaluz, N., Bhimraj, A., Guanci, M. M., Seder, D. B., & Singh, J. M. (2016). The insertion and management of external ventricular drains: An evidence-based consensus statement: A statement for healthcare professionals from the Neurocritical Care Society. *Neurocritical Care, 24*(1), 61–81. https://doi.org/10.1007/s12028-015-0224-8
15. Guidry, C. A., Mallicote, M. U., Petroze, R. T., Hranjec, T., Rosenberger, L. H., Davies, S. W., & Sawyer, R. G. (2014). Influence of bronchoscopy on the diagnosis of and outcomes from ventilator-associated pneumonia. *Surgical Infections, 15*(5), 527–532. https://doi.org/10.1089/sur.2013.142
16. Gupta, S., Heman-Ackah, S. E., & Lalwani, A. K. (2012). Are prophylactic antibiotics necessary for otologic surgery? *The Laryngoscope, 122*(6), 1183–1184. https://doi.org/10.1002/lary.23302
17. Hospenthal, D. R., Murray, C. K., Andersen, R. C., Bell, R. B., Calhoun, J. H., Cancio, L. C., Cho, J. M., Chung, K. K., Clasper, J. C., Colyer, M. H., Conger, N. G., Costanzo, G. P., Crouch, H. K., Curry, T. K., D'Avignon, L. C., Dorlac, W. C., Dunne, J. R., Eastridge, B. J., Ficke, J. R., . . . Wortmann, G. W. (2011). Guidelines for the prevention of infections associated with combat-related injuries: 2011 update: Endorsed by the Infectious Diseases Society of America and the Surgical Infection Society. *The Journal of Trauma: Injury, Infection, and Critical Care, 71*(2 Suppl. 2), S210–S234. https://doi.org/10.1097/TA.0b013e318227ac4b
18. Jones, B. E., Ramirez, J. A., Oren, E., Soni, N. J., Sullivan, L. R., Restrepo, M. I., Musher, D. M., Erstad, B. L., Pickens, C., Vaughn, V. M., Helgeson, S. A., Crothers, K., Metlay, J. P., Bissell Turpin, B. D., Cao, B., Chalmers, J. D., Dela Cruz, C. S., Gendlina, I., Hojat, L. S., . . . Wilson, K. (2026). Diagnosis and management of community-acquired pneumonia: An official American Thoracic Society clinical practice guideline. *American Journal of Respiratory and Critical Care Medicine, 212*(1), 24–44. https://doi.org/10.1164/rccm.202507-1692ST
19. Joung, M. K., Lee, J.-a., Moon, S.-y., Cheong, H. S., Joo, E.-J., Ha, Y.-E., Sohn, K. M., Chung, S. M., Suh, G. Y., Chung, D. R., Song, J.-H., & Peck, K. R. (2011). Impact of de-escalation therapy on clinical outcomes for intensive care unit-acquired pneumonia. *Critical Care, 15*(2), Article R79. https://doi.org/10.1186/cc10072
20. Kalil, A. C., Metersky, M. L., Klompas, M., Muscedere, J., Sweeney, D. A., Palmer, L. B., Napolitano, L. M., O'Grady, N. P., Bartlett, J. G., Carratalà, J., El Solh, A. A., Ewig, S., Fey, P. D., File, T. M., Jr., Restrepo, M. I., Roberts, J. A., Waterer, G. W., Cruse, P., Knight, S. L., & Brozek, J. L. (2016). Management of adults with hospital-acquired and ventilator-associated pneumonia: 2016 clinical practice guidelines by the Infectious Diseases Society of America and the American Thoracic Society. *Clinical Infectious Diseases, 63*(5), e61–e111. https://doi.org/10.1093/cid/ciw353
21. Kullar, R., Frisenda, J., & Nassif, P. S. (2018). The more the merrier? Should antibiotics be used for rhinoplasty and septorhinoplasty?—A review. *Plastic and Reconstructive Surgery—Global Open, 6*(10), Article e1972. https://doi.org/10.1097/GOX.0000000000001972
22. López Pereira, P., Díaz-Agero Pérez, C., López Fresneña, N., Las Heras Mosteiro, J., Palancar Cabrera, A., Rincón Carlavilla, Á. L., & Aranaz Andrés, J. M. (2017). Epidemiology of surgical site infection in a neurosurgery department. *British Journal of Neurosurgery, 31*(1), 10–15. https://doi.org/10.1080/02688697.2016.1260687

23. Metlay, J. P., Waterer, G. W., Long, A. C., Anzueto, A., Brozek, J., Crothers, K., Cooley, L. A., Dean, N. C., Fine, M. J., Flanders, S. A., Griffin, M. R., Metersky, M. L., Musher, D. M., Restrepo, M. I., & Whitney, C. G. (2019). Diagnosis and treatment of adults with community-acquired pneumonia: An official clinical practice guideline of the American Thoracic Society and Infectious Diseases Society of America. *American Journal of Respiratory and Critical Care Medicine, 200*(7), e45–e67. https://doi.org/10.1164/rccm.201908-1581ST
24. Nagy, Z., Szabó, D., Agócs, G., Szilágyi, K., Rojcsik, Z., Budai, J., Papp, Z., Padányi, C., Erőss, L., Sipos, L., & Banczerowski, P. (2025). Antibiotic prophylaxis in instrumented lumbar spine surgery: Cefazolin outperforms clindamycin regardless of duration. *Antibiotics, 14*(8), Article 830. https://doi.org/10.3390/antibiotics14080830
25. North American Spine Society. (2013). *Antibiotic prophylaxis in spine surgery: Evidence-based clinical guidelines for multidisciplinary spine care* (Revised ed.). https://www.spine.org/Portals/0/Assets/Downloads/ResearchClinicalCare/Guidelines/AntibioticProphylaxis.pdf
26. Patel, P. N., Jayawardena, A. D. L., Walden, R. L., Penn, E. B., & Francis, D. O. (2018). Evidence-based use of perioperative antibiotics in otolaryngology. *Otolaryngology–Head and Neck Surgery, 158*(5), 783–800. https://doi.org/10.1177/0194599817753610
27. Raman, K., Nailor, M. D., Nicolau, D. P., Aslanzadeh, J., Nadeau, M., & Kuti, J. L. (2013). Early antibiotic discontinuation in patients with clinically suspected ventilator-associated pneumonia and negative quantitative bronchoscopy cultures. *Critical Care Medicine, 41*(7), 1656–1663. https://doi.org/10.1097/CCM.0b013e318287f713
28. Saeed, O., Tribble, D. R., Biever, K. A., Crouch, H. K., & Kavanaugh, M. (2018). Infection prevention in combat-related injuries. *Military Medicine, 183*(Suppl. 2), 137–141. https://doi.org/10.1093/milmed/usy077
29. Sanabria, A., Valdivieso, E., Gomez, G., & Echeverry, G. (2006). Prophylactic antibiotics in chest trauma: A meta-analysis of high-quality studies. *World Journal of Surgery, 30*(10), 1843–1847. https://doi.org/10.1007/s00268-005-0672-y
30. Sharp, J. P., Magnotti, L. J., Weinberg, J. A., Swanson, J. M., Schoreppel, T. J., Clement, L. P., Wood, G. C., Fabian, T. C., & Croce, M. A. (2015). Adherence to an established diagnostic threshold for ventilator-associated pneumonia contributes to low false-negative rates in trauma patients. *Journal of Trauma and Acute Care Surgery, 78*(3), 468–474.
31. Tan, T., Lee, H., Huang, M. S., Rutges, J., Marion, T. E., Mathew, J., Fitzgerald, M., Gonzalvo, A., Hunn, M. K., Kwon, B. K., Dvorak, M. F., & Tee, J. (2020). Prophylactic postoperative measures to minimize surgical site infections in spine surgery: Systematic review and evidence summary. *The Spine Journal, 20*(3), 435–447. https://doi.org/10.1016/j.spinee.2019.09.013
32. Trubiano, J. A., Vogrin, S., Chua, K. Y. L., Bourke, J., Yun, J., Douglas, A., Stone, C. A., Yu, R., Groenendijk, L., Holmes, N. E., & Phillips, E. J. (2020). Development and validation of a penicillin allergy clinical decision rule. *JAMA Internal Medicine, 180*(5), 745–752. https://doi.org/10.1001/jamainternmed.2020.0403
33. UTHealth Houston. (2022, February). *Antibiotic therapy*. McGovern Medical School, Department of Surgery. https://med.uth.edu/surgery/antibiotic-therapy/
34. Vanderbilt University Medical Center, Division of Acute Care Surgery. (2025). *Infectious workup and antimicrobial stewardship guideline*. https://www.vumc.org/trauma-and-scc/sites/default/files/public_files/Protocols/Infectious-Workup-and-Antimicrobial-Stewardship-PMG-7.18.25.pdf
35. Younan, D., Delozier, S. J., Adamski, J., Loudon, A., Violette, A., Ustin, J., Tinkoff, G., Moorman, M. L., McQuay, N., & UHRISES Research Consortium. (2020). Factors predictive of ventilator-associated pneumonia in critically ill trauma patients. *World Journal of Surgery, 44*(4), 1121–1125. https://doi.org/10.1007/s00268-019-05286-3